Inference Rules

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Review

A **truth assignment** assigns $\mathbf{T}$ or $\mathbf{F}$ to each propositional variable.
A **tautology** is a propositional form that is true under every truth assignment.
Two propositional forms A and B are **logically equivalent**, written $A \equiv B$, if they have the same truth value under every truth assignment.
Truth tables allow us to check these claims by considering all possible truth assignments.

De Morgan's laws

For propositions P and Q, **De Morgan's laws** are

$$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$$

and

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q).$$

The first says that "neither P nor Q" means that both P and Q are false.
The second says that "not both" means that at least one of P and Q is false.

Checking the first law

To establish

$$\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q),$$

compare the two expressions under every truth assignment.

P	Q	$P \vee Q$	$\neg(P \vee Q)$	$(\neg P) \wedge (\neg Q)$
T	**T**	**T**	**F**	**F**
T	**F**	**T**	**F**	**F**
F	**T**	**T**	**F**	**F**
F	**F**	**F**	**T**	**T**

The last two columns are identical, so the forms are logically equivalent.

Proof by exhaustion

A **proof by exhaustion** divides a claim into finitely many cases and verifies the claim in every case.
The truth-table argument for De Morgan's law has four cases:

$$(\mathbf{T},\mathbf{T}),\ (\mathbf{T},\mathbf{F}),\ (\mathbf{F},\mathbf{T}),\ (\mathbf{F},\mathbf{F}).$$

Because these are all possible truth assignments for P and Q, no additional case remains.
Exhaustion proves a claim only when the listed cases are complete.

Checking the second law

Consider

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q).$$

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$(\neg P) \vee (\neg Q)$
T	**T**	**T**	**F**	**F**
T	**F**	**F**	**T**	**T**
F	**T**	**F**	**T**	**T**
F	**F**	**F**	**T**	**T**

The last two columns agree in all four cases.

A finite counting argument

Claim. If three socks are selected from a drawer containing only black and white socks, at least two of the selected socks have the same color.
Proof. Let b be the number of selected black socks. Then

$$b \in \{0, 1, 2, 3\}.$$

If $b = 0$ or $b = 1$, at least two selected socks are white.
If $b = 2$ or $b = 3$, at least two selected socks are black.
These four cases exhaust the possibilities, so the selection always contains two socks of the same color.

Limits of exhaustion

A truth table for n propositional variables has 2^n rows.
Thus a direct truth-table proof grows quickly:

n	2	5	10	20	30
2^n	4	32	1024	1,048,576	1,073,741,824

For larger arguments, it is more useful to combine smaller valid steps than to enumerate every truth assignment.

A deduction

Consider the statements:

- It is raining.
- If it is raining, then the road is wet.
- If the road is wet, then driving very fast is dangerous.

If all three statements are true, we may conclude:

Driving very fast is dangerous.

The conclusion is obtained by two short logical steps.

Symbolic form

Define the propositions

$$\begin{aligned} R &= \text{“It is raining,”} \\ W &= \text{“The road is wet,”} \\ D &= \text{“Driving very fast is dangerous.”} \end{aligned}$$

The three given statements are

$$R, \qquad R \Rightarrow W, \qquad W \Rightarrow D.$$

The proposed conclusion is D.

Arguments

An **argument** consists of one or more **premises** and a **conclusion**.

- The premises are the propositions from which we reason.
- The conclusion is the proposition claimed to follow from them.

Example

Premise: It is raining and the wind is blowing.

Conclusion: Therefore, it is raining.

The argument is the whole premise–conclusion combination, not the conclusion alone. Calling it an argument does not yet say it is valid.

Argument forms

For the previous example, let

$$P = \text{"It is raining"}, \qquad Q = \text{"The wind is blowing"}.$$

The premise is $P \wedge Q$ and the conclusion is P. We write $\therefore$, read "therefore," before the conclusion:

$$\underbrace{P \wedge Q}_{\text{premise}} \quad \therefore \quad \underbrace{P}_{\text{conclusion}} \quad .$$

An **argument form** is the logical structure expressed using propositional variables and logical connectives.
Now let P, Q stand for any propositions. The same form describes: "12 is even and positive. Therefore, 12 is even."
Different statements can have the same logical structure.

Comparing two argument forms

Let P mean "It is raining" and Q mean "The wind is blowing."
Argument 1

It is raining **and** the wind is blowing. Therefore, it is raining.

$$P \wedge Q \quad \therefore P.$$

If the premise is true, the conclusion must be true.

Argument 2

It is raining **or** the wind is blowing. Therefore, it is raining.

$$P \vee Q \quad \therefore P.$$

The wind could be blowing without rain: a true premise and a false conclusion.
In propositional logic, we test the form across all truth assignments, rather than checking today's weather.

Valid and invalid arguments

An argument form is **valid** if, under every truth assignment, whenever all its premises are true, its conclusion is also true.
Equivalently, a valid argument has no truth assignment for which

all premises are true   and   the conclusion is false.

An argument is **invalid** if such a truth assignment exists. This truth assignment is called a **counterexample** to validity.

To establish invalidity, one counterexample is enough.
To establish validity, we must show that it is impossible for all premises to be true while the conclusion is false.

Validity and truth are different

Valid does not mean that the conclusion is always true.
Invalid does not mean that the conclusion is always false.

Under the actual truth values of its propositions, an argument can have any of the following combinations:

	conclusion true	conclusion false
valid	possible	possible
invalid	possible	possible

If a valid argument has a false conclusion, at least one of its premises must be false.
Validity asks whether true premises can lead to a false conclusion; it does not evaluate the conclusion by itself.

Two valid arguments

Define the propositions

$$P = \text{“12 is even”}, \qquad Q = \text{“12 is greater than 10”},$$
$$R = \text{“11 is even”}, \qquad S = \text{“11 is greater than 10”}.$$

True conclusion
12 is even and greater than 10.
Therefore, 12 is even.

$$P \wedge Q \qquad \therefore P.$$

The premise and conclusion are true.

False conclusion
11 is even and greater than 10.
Therefore, 11 is even.

$$R \wedge S \qquad \therefore R.$$

The premise and conclusion are false.

Both have the form

$$A \wedge B \qquad \therefore A.$$

This form is valid: if $A \wedge B$ is true, then its conjunct A must be true.

Two invalid arguments

Recall that P, Q concern 12 and R, S concern 11:

$$P = \mathbf{T}, \quad Q = \mathbf{T}, \qquad R = \mathbf{F}, \quad S = \mathbf{T}.$$

True conclusion
12 is even or greater than 10.
Therefore, 12 is even.

$$P \vee Q \qquad \therefore P.$$

The premise and conclusion are true.

False conclusion
11 is even or greater than 10.
Therefore, 11 is even.

$$R \vee S \qquad \therefore R.$$

The premise is true and the conclusion is false.

Both have the form

$$A \vee B \qquad \therefore A.$$

This form is invalid: $A = \mathbf{F}, B = \mathbf{T}$ makes the premise true and the conclusion false. The second argument is a concrete counterexample.

Why validity matters

A valid argument does not guarantee that its premises are true.
It guarantees that if all the premises are true, the conclusion cannot be false:

$$\boxed{\text{true premises} + \text{valid argument} \quad \Longrightarrow \quad \text{true conclusion}}.$$

Therefore, deductive reasoning has two responsibilities:

1. justify the premises;
2. move from them to the conclusion using valid arguments.

Valid reasoning preserves truth from the premises to the conclusion.

Rules of inference

A deduction is built by connecting small valid arguments.
The same valid argument forms occur repeatedly. A **rule of inference** is a valid argument form used as one step in a deduction.
Once a rule has been shown to be valid, we may reuse it without checking every truth assignment again.
We will learn several common rules of inference and use them to construct deductions one step at a time.
We begin with a rule involving implication: **modus ponens**.

Modus ponens

The rule **modus ponens** is

$$\frac{P \qquad P \Rightarrow Q}{Q}.$$

The expressions above the line are the premises; the expression below the line is the conclusion.
The premise $P \Rightarrow Q$ rules out exactly the case

$$P = \mathbf{T}, \qquad Q = \mathbf{F}.$$

If we also know

$$P = \mathbf{T}, \qquad (P \Rightarrow Q) = \mathbf{T},$$

then $Q = \mathbf{F}$ is impossible. Therefore $Q = \mathbf{T}$.

The rain argument

Recall the premises

$$R, \qquad R \Rightarrow W, \qquad W \Rightarrow D.$$

From R and $R \Rightarrow W$, modus ponens gives W.
From W and $W \Rightarrow D$, modus ponens gives D.
Therefore D logically follows from the three premises.

Modus tollens

The rule **modus tollens** is

$$\begin{array}{l} P \Rightarrow Q \\ \neg Q \\ \hline \neg P \end{array}$$

This rule uses the contrapositive: $P \Rightarrow Q$ is logically equivalent to $\neg Q \Rightarrow \neg P$.
If Q is false, then P cannot be true without making $P \Rightarrow Q$ false.

Hypothetical syllogism

The rule **hypothetical syllogism** is

$$\begin{array}{c} P \Rightarrow Q \\ Q \Rightarrow R \\ \hline P \Rightarrow R \end{array}$$

If P is true, the first premise gives Q.
Once Q is true, the second premise gives R.
Hence P being true guarantees R.

Disjunctive syllogism

The rule **disjunctive syllogism** is

$$\begin{array}{l} P \vee Q \\ \neg P \\ \hline Q \end{array}$$

The first premise says that at least one of P and Q is true.
The second premise rules out P, so Q must be true.

Three basic rules

Addition

$$\frac{P}{P \vee Q}$$

Simplification

$$\frac{P \wedge Q}{P}$$

Conjunction

$$\frac{\begin{array}{l} P \\ Q \end{array}}{P \wedge Q}$$

Each rule is valid for every truth assignment of P and Q.

A deduction using two rules

Premises:

$$P \Rightarrow Q, \qquad P \vee R, \qquad \neg R.$$

Conclusion: Q.

statement	reason
$P \vee R$	premise
$\neg R$	premise
P	disjunctive syllogism
$P \Rightarrow Q$	premise
Q	modus ponens

Is this argument valid?

Let P, Q, R be propositional variables. Consider the argument

$$P \Rightarrow R, \qquad Q \Rightarrow R \qquad \therefore \qquad (P \vee Q) \Rightarrow R.$$

The question is whether the two premises being true guarantees that the conclusion is true.
No single rule of inference introduced so far has exactly this form. We will instead replace expressions by logically equivalent forms.

Equivalences for the deduction

For propositional variables A, B, C, recall

$$A \Rightarrow B \equiv (\neg A) \vee B \qquad \text{and} \qquad \neg(A \vee B) \equiv (\neg A) \wedge (\neg B).$$

The two **distributive laws** are

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C),$$
$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C).$$

The next deduction uses the second distributive law from right to left:

$$(A \vee B) \wedge (A \vee C) \equiv A \vee (B \wedge C).$$

Rewriting the premises

By the conjunction rule, the two premises give

$$(P \Rightarrow R) \wedge (Q \Rightarrow R).$$

Now replace each form by a logically equivalent form:

$$\begin{aligned}
&(P \Rightarrow R) \wedge (Q \Rightarrow R) && \\
&\quad \equiv (\neg P \vee R) \wedge (\neg Q \vee R) && \text{implication equivalence} \\
&\quad \equiv (\neg P \wedge \neg Q) \vee R && \text{distributive law} \\
&\quad \equiv \neg(P \vee Q) \vee R && \text{De Morgan's law} \\
&\quad \equiv (P \vee Q) \Rightarrow R && \text{implication equivalence.}
\end{aligned}$$

Therefore $(P \vee Q) \Rightarrow R$ follows from the two premises.

Summary

- A proof by exhaustion verifies a complete finite list of cases.
- An argument is valid when true premises cannot have a false conclusion.
- A rule of inference is a valid argument form used as a deduction step.
- Modus ponens, modus tollens, and the two syllogisms support short, checkable deductions.
- Logical equivalences allow one propositional form to be replaced by another with the same truth values.